Annexure F – L Huson & Associates Pty Ltd Review Report

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13 April 2016

WestWind Energy Pty Ltd
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Attention: Adam Gray

Dear Adam

LAL LAL WIND FARM
L. HUSON & ASSOCIATES PTY LTD REVIEW REPORT

INTRODUCTION

Marshall Day Acoustics Pty Ltd (MDA) prepared a report in relation to a proposed planning permit amendment for the Lal Lal Wind Farm titled Rp 002 2015386ML - *Lal Lal Wind Farm - NZS 6808:2010 Noise Assessment* dated 11 September 2015 (the MDA Report).

The current planning permit for the wind farm, reference PL-SP/05/0461, was issued in 2009 and refers to NZS 6808:1998 *Acoustics – The assessment and measurement of sound from wind turbine generators* (NZS 6808:1998) which was the current version of the standard at that time and which was referenced in the relevant Victorian Planning Provisions and Victorian Government's *Policy and planning guidelines for development of wind energy facilities in Victoria* (the Victorian Guidelines) dated May 2003. The MDA Report provides an updated noise assessment for the wind farm using the revised, 2010 version of the New Zealand Standard, 6808:2010 *Acoustics – Wind farm noise* (NZS 6808:2010) as this standard is nominated across all recent iterations of the Victorian Guidelines.

The MDA Report was incorporated into an application package prepared by WestWind Energy Pty Ltd which seeks approval for several proposed amendments to the current planning permit. The proposed amendments relevant to noise include updating references from NZS 6808:1998 to NZS 6808:2010, and proposed changes to the permit's conditions relating to post-construction noise commissioning requirements.

A peer review of the MDA Report has been prepared by L Huson & Associates Pty Ltd. The review forms part of a submission to the planning permit amendment by the Lal Lal Environment Protection Association Inc. The findings of the peer review are detailed in a report titled *Review – Lal Lal Wind Farm Planning Permit Amendment October 2015* and dated December 2015 (the Peer Review Report).

The Peer Review Report seems to generally support the proposal to amend the planning permit for the Lal Lal Wind Farm, at least to the extent noted in its introduction:

> *We support a revision to the current noise conditions contained in planning permit number PL-SP/05/0461, dated 30 April 2009, for the Lal Lal Wind Farm to reflect recent planning guidelines that refer to the 2010 version of the New Zealand Standard 6808:2010 'Acoustics – Wind farm noise'.*

With regard to the application of NZS 6808:2010 as documented in the MDA Report, the Peer Review Report also acknowledges that:

> [...] *the assessment by MDA has applied an assessment in accordance with NZS6808:2010* [...].

However, the Peer Review Report does not support all of the proposed changes to the planning permit including in the WestWind Energy application.

ISO 9001

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The Peer Review Report does provide comments regarding a number of technical aspects of the assessment detailed in the MDA Report. Key technical aspects have been broadly categorised by MDA as follows:

- Noise sensitive locations
  Identifying noise sensitive locations neighbouring the proposed wind farm
- Noise limits
  Nominating suitable noise limits for proposed wind farm neighbours including host landowner properties
- Turbine sound data
  Correctly accounting for the likely wind turbine sound emissions
- Noise modelling
  Implementing a suitable noise model, based on ISO 9613-2:1996[1]

MDA has considered these aspects raised in the Peer Review Report. Our comments about each of the above four topics are itemised separately in this letter along with a brief overview of the various iterations of the Victorian Government's *Policy and planning guidelines for development of wind energy facilities in Victoria* which are topical to the wind farm.

VICTORIAN GUIDELINES

Several iterations of the Victorian Guidelines have been relevant to the Lal Lal Wind Farm during the various phases of its planning. For clarity, relevant versions of the Victorian guideline document are detailed in Table 1 below along with notes about the corresponding stage of the wind farm's planning.

Table 1: Evolution of the Victorian guideline's noise related components

Document iteration	Noise related items	Wind farm stage
May 2003	Reference to NZS 6808:1998	Applicable at the time of the panel hearing
June 2015	Reference to NZS 6808:2010 Details the concept of High Amenity Areas	Applicable at the time the MDA Report was prepared
November 2015	No significant changes relevant to noise	Applicable at the time the Peer Review Report was prepared
January 2016	Provides additional guidance regarding High Amenity Areas, as discussed in the *NOISE LIMIT* section below	Currently applicable

[1] ISO 9613-2:1996 *Acoustics – Attenuation of sound during propagation outdoors Part 2: General method of calculation* (ISO 9613-2:1996)

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NOISE SENSITIVE LOCATIONS

The Peer Review Report lists two sets of coordinates that have not been included in the MDA Report. The two properties are detailed in Table 2.

Table 2: Properties identified in the Peer Review Report as missing in the MDA Report (GDA94)

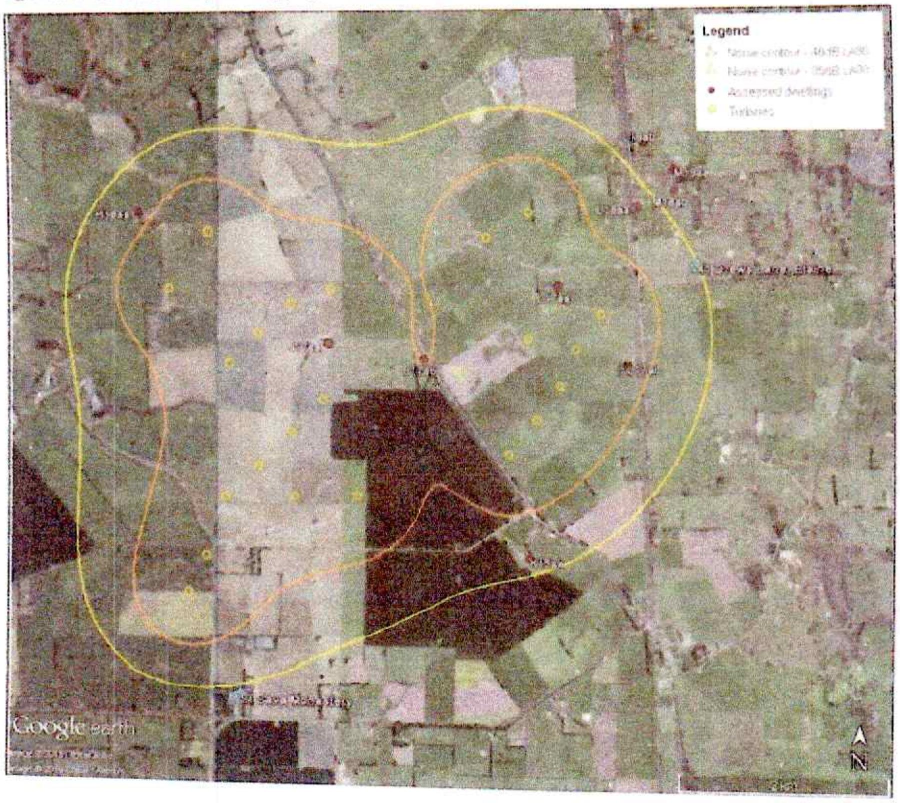
Reference	Easting	Northing	Zone*
41 Drews Lane, Elaine	238,476	5,818,187	55H
St Sava Monastery	762,709	5,814,190 **	54H

* The coordinate reference and zoning are not detailed in the Peer Review Report. The reference system and zones noted in this table are based on current site understanding.

** The Northing coordinate listed in the Peer Review Report is *581419.5*. This value may be a typo and has been interpreted as '5,814,190' for the current review.

The properties presented in Table 2 are also shown in Figure 1 together with the 35 and 40 dB L_{A90} noise contours[1].

Figure 1: Noise contour map showing properties listed in the Peer Review Report



[1] As provided in Section E2 of Appendix E of the MDA Report

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41 Drews Lane, Elaine

WestWind Energy has advised that the dwelling at 41 Drews Lane, Elaine did not exist at the time when the original planning permit was issued[3]. This dwelling is therefore not considered a relevant noise sensitive location in accordance with Condition 23 of the original planning permit which states:

> [...] *the operation of the wind energy facility must comply with the noise criteria specified in NZS6808:1998 [...] at any dwelling existing on land in the vicinity of the proposed wind energy facility as at the date of the issue of this permit, to the satisfaction of the Minister for Planning.*

Notwithstanding this, as indicated in Figure 1, noise levels from the Lal Lal Wind Farm at this dwelling are predicted to be 35 dB L_{A90} in accordance with the methodology detailed in Section 7.0 of the MDA Report.

Compliance with a base noise limit of 40 dB L_{A90} is predicted to be achieved at this new dwelling by a margin of 5 dB.

St Sava Monastery

The St Sava Monastery was referenced as property I14aa in the noise assessment[4] submitted as part of the original planning permit application. Section 7.0 of the 2008 MDA Report states:

> *One of the properties (I14aa) identified for background noise monitoring using the preliminary layout will not be included in the noise assessment as noise levels were predicted below 35dBA using the proposed layout.*

As indicated in the noise contour map provided in Figure 1, noise levels from the Lal Lal Wind Farm at the St Sava Monastery are predicted to be 34 dB L_{A90} in accordance with the methodology detailed in Section 7.0 of the MDA Report.

Compliance with a base noise limit of 40 dB L_{A90} is therefore predicted to be achieved at the St Sava Monastery by a margin of 6 dB.

NOISE LIMITS

The MDA Report applies a base noise limit of 40 dB L_{A90} for the noise assessment of the Lal Lal Wind Farm. The Peer Review Report provides the following comments about this issue:

> *The MDA REPORT has elected not to set a lower baseline target noise levels of 35 dB L_{A90} that would be suitable for a high amenity noise area.*
>
> *We disagree with the arguments in the JACOBS and MDA reports against allowing the area to be designated a high amenity noise area.*
>
> *[...]*
>
> *It would be appropriate to require an objective assessment of high amenity noise areas surrounding the Project.*

The concept of a high amenity area was introduced in the 2010 version of NZS 6808. Section 5.3 of NZS 6808:2010 provides details of high amenity noise limits, requiring that where a noise sensitive location is situated within a high amenity area, as defined in Sections 5.3.1 and 5.3.2 of NZS 6808:2010, wind farm noise levels (L_{A90}) during evening and nigh-time periods should not exceed the background noise level (L_{A90}) by more than 5 dB or 35 dB L_{A90}, whichever is the greater, for wind speeds below 6 m/s at hub height. High amenity noise limits are not applicable during the daytime period.

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[3] It is our understanding that the occupiers of the dwelling lodged an objection to the amendment which was subsequently withdrawn.

[4] Report No. 001 R01 2007344 *Lal Lal Wind Farm - Noise Assessment* dated 5 February 2008 (the 2008 MDA Report)

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Section 5.3 of NZS 6808:2010 states that:

> *A high amenity noise limit should be considered where a plan promotes a higher degree of protection of amenity related to the sound environment of a particular area...*

Subsequent to this, Section C5.3.1 of the Standard provides a calculation procedure for objectively evaluating whether relevant background noise levels for a potential high amenity area are sufficiently low to confirm the application of a high amenity noise limit.

High amenity noise limits are referenced in the Victorian Guidelines. The comments provided in the guidelines about high amenity limits have been revised across several recent versions of the guidelines. The most recent version of the guidelines, dated January 2016, provides arguably the most comprehensive discussion of how to apply the concept of high amenity areas in the Victoria context. As the January 2016 guidelines post-date both the MDA Report and the Peer Review report, they are discussed in some detail here.

Section 5.1.2.(a) of the January 2016 Victorian Guidelines states the following.

> *Under section 5.3 of the Standard, a 'high amenity noise limit' of 35 decibels applies in special circumstances. All wind farm applications must be assessed using section 5.3 of the Standard to determine whether a high amenity noise limit is justified for specific locations, following procedures outlined in clause C5.3.1 of the Standard. Guidance can be found on this issue in the VCAT determination for the Cherry Tree Wind Farm.*

The definition of a high amenity area provided in NZS 6808:2010 is specific to New Zealand planning legislation and guidelines. A degree of interpretation is therefore required when determining how to apply the concept of high amenity area in Victoria, specifically in order to satisfy Section 5.1.2.a of the guidelines.

As detailed above, the current version of the Victorian Guidelines recommends that guidance on the application of the High Amenity noise limit be sourced from the VCAT determination detailed in the Cherry Tree Wind Farm Pty Ltd v Mitchell Shire Council decisions[5] which provide some additional context to this issue.

Paragraph 53 of the Cherry Tree Wind Farm Decision states the following:

> *The Tribunal does not accept that the permit conditions need to refer to the High Amenity Area provisions of the New Zealand standard because it has not been established that any such area could reasonably be identified within the environs of this wind energy facility. [...]*

[5] Mitchell Shire Council interim decision dated 4 April 2013 (the Cherry Tree Wind Farm Interim Decision) and Mitchell Shire Council decision dated 27 November 2013 (the Cherry Tree Wind Farm Decision)

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Justification for this statement was provided in Paragraphs 107 to 109 of the Cherry Tree Wind Farm Interim Decision:

> 107. *We were invited by the respondents to treat the subject land and the locality as a high amenity area. This invitation meets with the immediate conundrum that the language of the standard is not translatable to the Victorian planning framework. The "plan" referred to in section 5.3 is a plan as defined by the Resources Management Act of New Zealand. Section 43AA of that Act defines "plan" to mean "a regional plan or a district plan". No such animals exist under the Victorian legislation.*
>
> 108. *Applying the standard mutatis mutandis to the Victorian experience we treat the plan referred to in the standard as a planning scheme approved under the Planning and Environment Act 1987. The Mitchell Planning Scheme does not anywhere expressly or by implication "promote a higher degree of protection of amenity related to the sound environment of a particular area". Approaching the matter by a process of elimination it can be seen with certainty that the controls contained within the Farming zone, which includes most of the locality, do not answer this description. The purpose of the Farming zone is to encourage agricultural use, which is not an inherently quiet land use. In fact reference to the zone purposes confirms that agricultural use is to be preferred to residential use if there is potential conflict between the two.*
>
> 109. *Accordingly the Tribunal concludes that the subject land and its locality is not capable of designation as a high amenity area because it does not possess the necessary characteristics of such an area as specified in the NZ standard.*

As detailed in Paragraph 108, for the land surrounding the wind farm to be considered a high amenity area, the zoning of the land must be identified in the relevant planning scheme as *promoting a higher degree of protection of amenity related to the sound environment*.

As discussed above, NZS 6808:2010 states that a high amenity noise limit should be considered where a plan promotes a higher degree of protection of amenity related to the sound environment of a particular area. The area surrounding the proposed wind farm is generally zoned Farming Zone as shown in the planning map presented in Appendix C of the MDA Report.

The following points are noted about the Farming Zone:

- The Farming Zone in the Moorabool Planning Scheme dated 4 February 2016 is not considered to promote a higher degree of protection of amenity related to the sound environment.
- The Victoria Planning Provisions Practice Note prepared by the Department of Sustainability and Environment titled *Applying the rural zones* and dated March 2007 states the following:

  The Farming Zone is designed to encourage diverse farming practices, some of which can have significant off-site impacts. For this reason, the level of amenity that can be expected in this zone will usually not be compatible with sensitive uses, particularly housing.

Based on the above and following guidance from the VCAT determination for the Cherry Tree Wind Farm, as recommended by the Victorian Guidelines, the high amenity noise limit detailed in NZS 6808:2010 is not considered to be applicable for residential properties in the vicinity of the Lal Lal Wind Farm.

As detailed in Section 6.3 of the MDA Report, the applicable base noise limit is considered to be 40 dB L_{A90} for all noise sensitive locations, as defined in NZS 6808:2010.

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Host noise limits

An increased base noise limit of 45 dB L_{A90} was presented in Section 6.3 of the MDA Report based on guidance from the final report by *The European Working Group on Noise from Wind Turbines* (ETSU-R-97).

This is consistent with the recommendation of the Draft National Guidelines[1] to increase the applicable base noise limit by 5 dB.

TURBINE SOUND DATA

The purpose of the noise assessment detailed in the MDA Report is to determine whether the proposed wind farm can achieve the requirements of the Victorian Guidelines and NZS 6808:2010 with a turbine model satisfying the requested specification amendments.

The MDA Report stated the following in Section 8.0 *Conclusion*:

> *An assessment has been undertaken, using the Senvion 3.2M114 wind turbine model with a hub height of 104m, in accordance with NZS 6808:2010 as required by the current Victorian Guidelines at twenty (20) residential properties identified by WestWind in the vicinity of the project.*
>
> [...]
>
> *If the turbine selection and/or layout are to be changed, compliance with the relevant noise limit will need to be reassessed.*

Sound Power Data

The sound power data used for predicted noise levels at residential properties in the vicinity of the proposed wind farm was provided by the manufacturer based on measurements undertaken in accordance IEC 61400-11, as recommended in Section 6.2.1 of NZS 6808:2010:

> *For the purposes of this Standard it is recommended that wind farm sound level predictions be based on the apparent sound power and tonality values for the nominated wind turbine model, determined in accordance with IEC 61400-11.*

A presentation[2] by Erik Sloth was also referenced in this section of the Peer Review Report to justify the claims regarding the inadequacy of sound power data determined in accordance with IEC 61400-11.

Reference to this presentation can be misleading. Some of the reasons are as follows:

- Quoting parts of a PowerPoint presentation out of the context of the oral presentation may misrepresent the intent of the author
- It is our understanding that the purpose of Erik Sloth's presentation was to critique the 2002 version of IEC 61400-11 which is now obsolete. In particular, we understand that the quoted text refers to the shortcomings of an assessment undertaken at 10 m Above Ground Level (AGL) and is therefore not necessarily relevant to a hub height assessment as required by NZS 6808:2010.

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[1] *EPHC National Wind Farm Development Guidelines – Draft dated July 2010*

[2] *Problems related to the use of the existing noise measurement standards when predicting noise from wind turbines and wind farms presented at the 2004 AusWind Conference*

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Tonality

The Peer Review Report provides the following comments regarding tonality:

> *A guarantee of compliance to earlier versions of IEC 61400-11, or the latest version with small sample size, is of little value if the assessment of tonality is "more akin to playing the lottery". We have no confidence that such a guarantee proposed in the MDA report will achieve the objective of ensuring the absence of tonal special audible characteristics in practice.*

As detailed in Section 3.4 of the MDA Report, NZS 6808:2010 emphasises the assessment of special audible characteristics including tonality during the post-construction measurement phase of a project:

> *[...] as special audible characteristics cannot always be predicted, consideration shall be given to whether there are any special audible characteristics of the wind farm sound when comparing measured levels with noise limits.*

An assessment of the potential for tonality is, however, possible during the planning phase of a project as detailed in Section 6.2.1 of NZS 6808:2010:

> *For the purposes of this Standard it is recommended that wind farm sound level predictions be based on the apparent sound power and tonality values for the nominated wind turbine model, determined in accordance with IEC 61400-11.*

An assessment of tonality in accordance with NZS 6808:2010 is included in the MDA Report based on tonality assessment results for the candidate turbine carried out according to IEC 61400-11:2006[3]. The outcome of the assessment, which was based on information provided by the manufacturer for the candidate turbine, was that a penalty for tonality was not considered applicable for any of the assessed wind speeds.

However, recognising that the constructed turbine model may differ from the candidate turbine and the ultimate objective is for the wind farm noise at receptor locations to be free of special audible characteristics (including tonality). With this in mind, Section 2.2.3 of the MDA Report states:

> *[...] we envisage that the procurement contract for the site would stipulate that the turbines must not produce emissions which would attract a penalty when assessed in accordance with the relevant noise criteria and any associated conditions of consent.*

[3] IEC 61400-11:2006 *Wind turbine generator systems – Part 11: Acoustic noise measurement techniques*

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NOISE MODELLING

ISO 9613-2:1996

The MDA Report applies ISO 9613-2:1996[ii] for predicting levels of noise from the Lal Lal Wind Farm.

This approach is considered consistent with NZS 6808:2010. In particular, Section 6.1.3 of NZS 6808:2010 states the following:

> *There is not a standardised sound propagation calculation method directly applicable to wind turbines. However, an example of a prediction method (from ISO 9613-2) that has been shown to correlate well with measured data for wind farms is detailed in Appendix D. This method provides a good balance between accuracy and completeness on one hand, and the effort of obtaining data to enter into the model on the other. Other prediction methods that can be shown to be appropriate for a given situation may be used, provided the details, assumptions, and limitations of the model are stated.*

Experience with relevant wind farm projects in jurisdictions referencing NZS 6808:2010, specifically Victoria and New Zealand, suggests that ISO 9613-2:1996 is the most commonly used method for predicting wind farm sound.

The use of ISO 9613-2:1996 for predicting wind farm noise (as detailed in Appendix D of the MDA report)is also supported by international research publications and measurement studies conducted by Marshall Day Acoustics. For these reasons, the ISO 9613-2:1996 prediction method is implemented by Marshall Day Acoustics for every wind farm project with noise predictions carried out in accordance with NZS 6808:2010

The Peer Review Report comments as follows regarding ISO 9613-2:1996:

> *The MDA report has used ISO9613 to predict noise levels surrounding the Project site. Reference has been made to a report by Bass, Bullmore and Sloth (1998) suggesting that this report "found that the ISO 9613-1:1996 model provided a robust representation of upper noise levels which may occur in practice, and provided a closer agreement between predicted and measured noise levels than alternative standards such as CONCAWE and ENM. Specifically, the report indicated the ISO 9613-2:1996 method generally tends to marginally over predict noise levels expected in practice." This statement is misleading. The Bass, Bullmore and Sloth report (Joule study) advised the use of an alternative model in preference to ISO9613.*

For reference, relevant extracts regarding CONCAWE, ENM and ISO 9613-2:1996 from the Conclusion of the Joule Report[iii] are noted here:

> *- Models that rely on analytical descriptions of sound propagation through the atmosphere [CONCAWE, ENM] are overly sensitive to changes in meteorological parameters. Variations in noise levels of up to 30dB(A) were predicted by these models, whereas measured variations under the same range of meteorological conditions were limited to less than 10dB(A).*
>
> *- The more advanced empirical model tested was that set out in ISO 9613-2. This method generally provide high levels of accuracy to within 2dB(A) in predicting received noise levels under 'conditions favourable to noise propagation', or downwind propagation.*

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ii International Standard ISO 9613-2:1996 Acoustics – *Attenuation of sound during propagation outdoors Part 2: General method of calculation* (ISO 9613-2:1996)

iii Bass, Bullmore and Sloth – *Development of a wind farm noise propagation prediction model*, Contract JOR3-CT95-0051, Final Report, January 1996 to May 1998.

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Further, page 143 of the Joule Report comments as follows regarding the accuracy of the prediction methods considered in that report and the selection of a preferred model:

> *The results of this section* [of the Joule Report] *have demonstrated that each of the ENM, IEA and ISO 9613 methods of calculation are capable of providing a high degree of accuracy when calculating far field noise levels radiated from elevated noise sources. The decision as to a recommended calculation procedure therefore rests on the simplicity of the models and the confidence limits that can be placed on their output.*

In light of these comments from the Joule Report, it is not clear why the Peer Review Report suggests that the quoted content from the MDA Report is misleading.

Terminology

As described in the preceding section, the MDA Report applies the ISO 9613-2:1996 prediction method. A note about terminology may be helpful at this stage.

In the strict sense, the MDA Report applies a modified version of ISO 9613-2:1996. That is, the ISO 9613-2:1996 method is applied in accordance with the standard with the exception of the attenuation factors for ground effect and screening which are adjusted in some cases as detailed in Appendix D of the MDA Report. Specifically:

- In instances where the ground terrain provides marginal or partial acoustic screening, the barrier effect is limited to not more than 2 dB
- Screening attenuation is calculated based on the screening expected for the source located at the tip height of the turbine (in contrast to hub height in non-adjusted ISO 9613 predictions)
- In instances where the ground falls away significantly between the source and receiver, such as valleys, an adjustment of 3 dB is added to the calculated sound pressure level. A terrain profile in which the ground falls away significantly is defined as one where the mean sound propagation height is at least 50 % greater than would occur over flat ground.

These adjustments are consistent with those proposed in the Joule Report and the Institute of Acoustics UK document *A good practice guide to the application of ETSU-R-97 for the assessment and rating of wind turbine noise*.

For simplicity of reference, the MDA Report refers to the adjusted model directly as ISO 9613-2:1996.

Ground factor

The ISO 9613-2:1996 prediction method accounts for sound absorption from the ground using a Ground Factor value which ranges from G=0, as would occur with concrete or asphalt, to G=1 which characterises porous ground which, according to the standard, "*...includes ground covered by grass, tress or other vegetation, and all other ground surfaces suitable for the growth of vegetation such as farming land.*"

Appendix D of the MDA Report provides a comprehension discussion justifying the choice of ISO 9613-2:1996 input parameters used for the noise assessment, including the application of a ground factor value of G=0.5.

The Peer Review Report states that the:

> *...MDA report references good practice guidelines from the UK for a justification to use G=0.5. Given the climate in the UK, it would be reasonable to use a 50/50 mixed ground terrain value of G=0.5 for an ISO9613 noise model in the UK. However, in Victoria this model input parameter is not considered appropriate. For example, the South Australian EPA wind farm noise models are required to use G=0 if ISO9613 is used.*

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Appendix D of the MDA Report provides several references that support the use of a ground factor value of G=0.5, among these is a UK Institute of Acoustics journal article dated March/April 2009 detailing a joint agreement between UK practitioners in the field of wind farm noise assessment. The joint statement indicates that the ISO 9613-2:1996 method is an appropriate standard for wind farm noise predictions and specifically designates G=0.5 as the appropriate ground characterisation. As noted by the Peer Review Report, the joint statement was prepared by UK practitioners and so would likely concern wind farm assessments in UK. The other references provided in Appendix D of the MDA Report are not specific to the UK and include guidance offered by NZS 6808:2010 and a study of wind farm noise from a Victorian wind farm.

The Peer Review Report comments noted above also reference the 2009 South Australian EPA Guidelines. A relevant extract from the South Australian guidelines is provided in Figure 2.

Figure 2: Extract from Page 10 of the 2009 South Australian EPA Guidelines

> A conservative approach should be used for predicting wind farm noise by calculating noise levels in octave bands from at least 63 to 4,000Hz to determine an overall predicted level and using the following inputs:
>
> - atmospheric conditions at 10°C and 80% humidity,
> - weather category 6 (if CONCAWE method is utilised),
> - hard ground (zero ground factor).
>
> If another prediction method and modelling inputs are employed to carrying out the noise level prediction, the details of the model should be clearly stated and the approach discussed with the Authority.

The use of a ground factor value of G=0.5 has been applied by MDA for several South Australian wind farm projects, with supporting discussions provided that are similar to those from Appendix D of the MDA Report. This approach has been accepted by the South Australian EPA for multiple projects[11] as an acceptable input parameter in conjunction with the other input parameters described in the MDA Report.

PANEL REPORT FOR THE LAL LAL WIND FARM 2009

The Peer Review Report includes a discussion of recommendations made by the Lal Lal Wind Farm panel hearing as detailed in the Panel Report titled *Lal Lal Wind Energy Facility Permit Application PL SP/05/0461 And Native Vegetation Removal Permit Application PL07/067* dated 11 February 2009.

The relevance of this Panel Report in relation to this planning permit amendment is limited as the relevant noise standard at the time was the 1998 version of NZS 6808 and not the current NZS 6808:2010.

Furthermore, the Panel Report extracts quoted in the Peer Review Report do not contradict the outcome of the MDA report.

Yours faithfully

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Christophe Delaire

Associate

[11] Such projects include Waterloo 2, Stony Gap, Ceres and Allendale